



451 Research Market Insight Report Reprint

Nozomi Networks deepens OT portfolio in push to secure critical infrastructure

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As Nozomi celebrates its 10th anniversary, recent funding lends weight to its foray into critical infrastructure security. With criminal and nation-state cyberthreats to industrial enterprises and critical infrastructure on the rise, the demand for operational technology security is growing.

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Introduction

Earlier this year, Nozomi Networks raised \$100 million in a series E funding round backed by Mitsubishi Electric Corp. and Schneider Electric SE. As Nozomi celebrates its 10th anniversary, that funding lends weight to its foray into critical infrastructure security. With criminal and nation-state cyberthreats to industrial enterprises and critical infrastructure on the rise, the demand for operational technology security is growing.

The need to bolster critical infrastructure against cyberattacks is underpinned by a similar push by Nozomi's competitor Dragos, as well as the original equipment manufacturers (OEMs) backing Nozomi's series E funding round, one of which, Schneider Electric, recently launched its own managed security service offering.

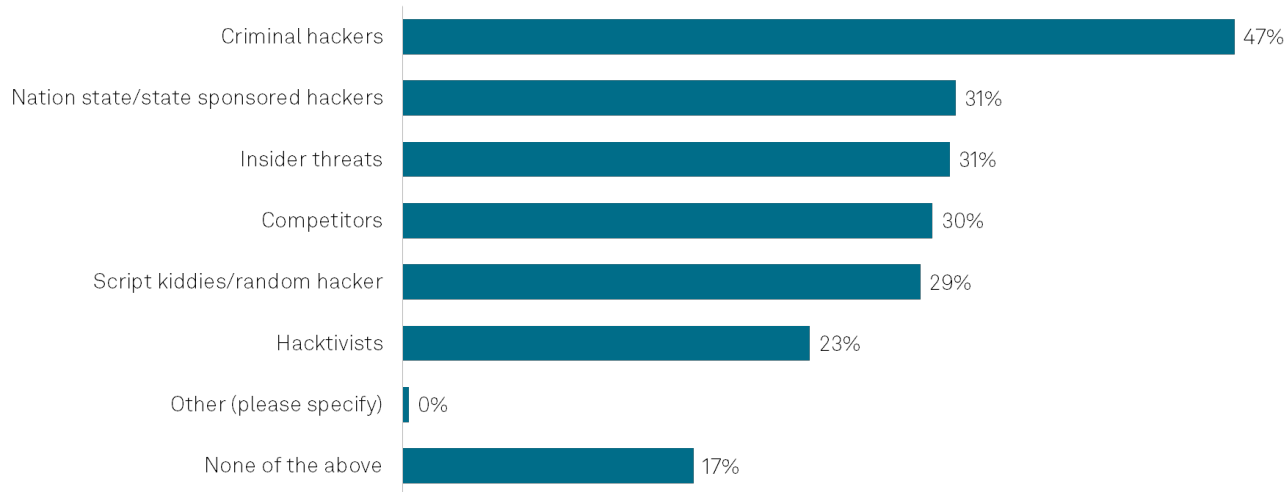
THE TAKE

Nozomi is often regarded as one of the “big four” in the segment of OT security, alongside other early incumbents Dragos, Claroty and Tenable (through its pickup of Indegy). These four are going through similar growth cycles and expanding their portfolios, with Dragos putting more focus into its threat intelligence and Claroty diversifying more into the “internet of any things,” including healthcare devices. Nozomi is deepening and widening its OT portfolio and extending its footprint in internet of things with a focus on AI-enabled approaches. Traditionally, Nozomi has a strong footprint in the oil & gas industry, and Dragos has a stronger customer base in the (power) utility sector, but with the backing of some notable OEMs, Nozomi is catching up.

Context

Power utilities, as part of critical national infrastructure, are becoming a top priority for all vendors, with the changing global balance of power and geopolitical tensions leading to security risks to critical infrastructure. According to 451 Research's Voice of the Enterprise: Internet of Things, Connectivity & Security 2023 survey, most enterprises still see criminal hackers as their primary digital adversary, but roughly one-third are well aware of nation-state-sponsored threats.

Most likely bad actors in OT and/or IT implementation attacks



Source: 451 Research's Voice of the Enterprise: Internet of Things, Connectivity & Security 2023.

Q. Which type of cybersecurity “bad actor” has attacked, or do you expect is most likely to attack, your operational technology and/or IoT implementations? Select all that apply.

Base: All respondents (n=302).

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San Francisco-based Nozomi Networks was founded in 2013 by Andrea Carcano and Moreno Carcullo. The \$100 million series E funding round brings Nozomi's total investment to \$250 million. Earlier funding includes a \$100 million series D round in 2021 led by Triangle Peak Partners with the participation of another OEM, Johnson Controls. Other investors include Activate Capital Partners, Energize Capital, Forward Investments, Keysight Technologies, Lux Capital, Notable Capital (fka GGV Capital), Planven Investments, Porsche Ventures and Samsung Ventures.

Nozomi has about 260 employees, roughly double the number since we last covered it in 2020. Aside from its headquarters in San Francisco, it has European headquarters in Switzerland and a presence in 20 countries including Australia, Brazil, Canada, Germany, Italy, Mexico, the Netherlands, Portugal, Singapore, Spain, United Arab Emirates and the UK. According to Nozomi, it now protects over 105 million OT, IoT and IT devices worldwide, with more than 12,000 installations across six continents. The company reports fivefold organic growth in annual recurring revenue since launching its flagship Vantage product in 2021.

Strategy

Nozomi has built a strong partner network of OEMs for critical infrastructure, or critical national infrastructure. These include partnerships with ABB Ltd., Emerson Electric Co., GE Vernova Inc., Hitachi Ltd., Johnson Controls, Mitsubishi Electric, Schneider Electric, Siemens AG and Yokogawa Electric Corp. The company increasingly works with managed security service providers (MSSPs) as partners, including Schneider Electric's managed security services and DeepSeas. It also services smaller utilities through MSSPs. Nozomi also partners with a community of global systems integrators, including Accenture PLC, Atos SE, BT, Deloitte, EY, IBM Security, Google Mandiant, NTT Data, Orange Cyberdefense, Telefónica SA and Thales SA, to address the strategic transformation requirements of enterprise CISOs.

Technology

Nozomi's 10-year journey mirrors the evolution of the OT security space, as well as the security journeys of many enterprises. The company's initial use case was asset detection because its customers were largely unaware of which connected devices were operational in their OT environments. Nozomi launched its first product, a network sensor providing passive detection, named Guardian (previously SCADAguardian); the next step was to add an on-premises central management console to provide a centralized view of the OT environment. Nozomi then added an active polling capability, and in early 2020 began to offer its customers asset and threat intelligence services through two optional subscription feeds. The next evolution of its product was the move from an on-premises offering to an enterprise-wide cloud offering, Vantage (early 2021), and expanding the threat intelligence with incident response playbooks.

As OT security increasingly becomes an enterprise risk, Nozomi sees a change in responsibilities. Where operations used to be responsible for OT security, the company sees a transfer of responsibility toward CISO-led programs, integrating responsibilities for both IT and OT security programs. To serve the CISO needs, Nozomi has added AI-powered analysis and response and compliance reporting. The company believes that as it gains asset intelligence from more customers, AI will accelerate its detection capabilities. The latest additions to its product portfolio are its new endpoint sensors, which install in Windows, Linux and MacOS environments to capture data and insight about key assets and network endpoints. Earlier this year, Nozomi launched Guardian Air, a wireless spectrum sensor built for OT and IoT environments. Nozomi builds its own wireless sensors and integrates them with the Nozomi platform to provide visibility for wireless connected sensors, devices, laptops and cell phones.

Competition

Nozomi is among the early entrants into the market of industrial cybersecurity, which gave it a good start compared with other vendors that have since entered the market. Nozomi vies with other early entrants offering industrial control system security solutions and asset visibility, including intrusion-detection offerings such as Armis, Axonius, Claroty, Dragos, Pas Global, Radiflow, SCADAfence (acquired by Honeywell International Inc. in 2023), SparkCognition, Tenable and XM-Cyber. Of these, Dragos and Claroty are Nozomi’s prime competitors and equally push into critical national infrastructure, with Dragos securing \$74 million of funding in 2023 and Claroty also securing a \$100 million round early in 2024. With its latest addition of Guardian Air, Nozomi primarily competes with Bastille in the wireless security segment. With its push toward securing programmable logic controllers and Level 1 devices, Nozomi will most likely run into NanoLock Security.

During the last several years, more traditional security vendors have joined the pure-play OT security vendors to extend their offerings to industrial systems, mostly through acquisitions, including Cisco Systems Inc., IBM Corp., Palo Alto Networks Inc., Trend Micro Inc., Sentinel One and Google (through its purchase of Mandiant). Additional competition comes from newer entries to the market like Eclipsium, Firedome, Sternum and VDOO (acquired by JFrog in 2021), which focus on embedded IoT and OT endpoint protection, and vendors like Darktrace, Karamba Security, Lansweeper, Netrise, Otorio, Perygee and Sepio.

SWOT Analysis

STRENGTHS By moving down to Level 1 devices and adding wireless security with Guardian Air, Nozomi aims to provide one of the most comprehensive OT security suites available in the market.	WEAKNESSES With the added endpoint sensors, Nozomi is pushing deeper into the OT technology stack but falls short on embedded security to cover all Purdue levels.
OPPORTUNITIES With a fragmented security landscape and enterprises dealing with multiple security vendors, Nozomi may have an opportunity to become a leading vendor as it increasingly covers the breadth of security challenges.	THREATS Nozomi’s competitor Dragos recently made a strong move to solidify its presence in utilities with its Community Defense Program, which gives small utilities free access to the Dragos Platform.

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